GEL Estimators as Projections Under Misspecification

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Abstract

This paper addresses an important problem in econometric analysis under misspecification: how should we interpret the pseudo-true values of the parameters? It is shown for the generalized empirical likelihood class of estimators that the pseudo-true value has an interpretation as a projection, however standard approaches to estimation will fail to be consistent for this parameter in general. An adjusted procedure is developed and it is shown that the estimators are consistent and have a limit distribution, which is Gaussian under correct specification and potentially non-Gaussian when the misspecification is severe.

1 Introduction

Economic models are frequently formulated as a set of moment conditions $g(X_i, \theta)$ which, for some value of the parameters θ_0, are mean zero: $\mathbb{E}\, g(X_i, \theta_0) = 0$. When the dimension of g is equal to the dimension of θ, existence of such a (possibly not-unique) θ_0 will generally hold as long as the distribution and support of X_i and the parameter space for Θ do not rule it out. When the dimension of g exceeds θ, then even for well-behaving models there might not exist a parameter value under which the moments are mean zero. In such a case, the model is said to be misspecified.

If we estimate the parameter θ by minimizing a criterion function, we can interpret the target parameter as the minimizer of the population criterion when the model is misspecified. This approach goes back to White (1982) in which the maximum likelihood estimator (MLE) is shown to be consistent for the parameter which minimizes the quasi-log-likelihood function. The interpretation in this case is that maximum likelihood can tell us which member of a parametric family is closest to the true data

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generating process, in a minimum Kullback-Leibler (KL) divergence sense. The MLE is generally still asymptotically normal, and hypothesis tests and confidence intervals can be designed which are robust to misspecification, in the sense that correct size and coverage can be obtained for pseudo-true values, regardless of whether it is known whether the model is misspecified.

In models defined or estimated by moment conditions instead of likelihoods, the empirical likelihood (EL) procedure has the attractive interpretation of estimating the model parameters θ using a nonparametric analogue of maximum likelihood. Estimating θ via EL involves solving an optimization problem, optimizing over both the parameter space and the set of discrete probability measures under which the moment conditions are satisfied. In this case, we would hope that under misspecification the solution in the population can also have the KL projection interpretation. In general this is not the case.

This paper outlines the properties of the pseudo-true value in the class of generalized empirical likelihood (GEL) estimators. Contrary to popular belief, we show that the standard implementation of GEL will generally fail to recover pseudo-true parameters which have the projection interpretation. We provide a simple modification which leads to recovery of the projection parameters and show that they can be consistently estimated. Finally, we derive the asymptotic distribution, which will fail to be Gaussian precisely when the standard implementation of GEL does not have a projection interpretation.

One interesting application of misspecification robust inference for GEL is using the implied probabilities as survey weights, as in Hellerstein and Imbens (1999), or as weights for the bootstrap as in Brown and Newey (2002). In both cases it is critical that the weights are constructed to satisfy the model constraints. When the model is sufficiently misspecified, standard methods of obtaining the weights will no longer satisfy the model constraints.

Our results on existence of a pseudo-true value are known in the convex analysis literature for fixed θ, and it is with minor modification that we adapt them to the setting most common in econometrics. The key insight from this literature is that the duality relationships commonly utilized for estimating these models can fail in the population. This failure of standard intuition on optimization problems comes from the fact that the nuisance parameter space in the primal problem is L^1, which is not reflexive. In Borwein and Lewis (1991) it is shown that a key requirement for strong duality in this infinite-dimensional setting is that the solution in the dual problem is in the relative interior of the domain for the dual problem. For sufficiently severe misspecification, the solution can lie on the boundary. In Borwein and Lewis (1993) and Csiszár (1995), it is shown that the primal problem can be extended, and the solution has the desired projection interpretation.

This is not the first paper to investigate the GEL setting under misspecification. It was established in Schennach (2007) that the EL estimator is generally not $\sqrt{n}$-consistent for any parameter value under misspecification when the moment conditions are unbounded. The possibility of similar problems for the exponential tilting estimator (ET) were discussed in Sueishi (2013). Recently, Schennach and Wahlstrom (2024) develop further a class of estimators which produce $\sqrt{n}$-consistent estimators of a pseudo-true parameter under model misspecification and unbounded moment restrictions. Rather than exploring alternative pseudo-true parameters, we focus on when the parameters targeted by GEL can have the attractive projection interpretation.

The rest of the paper proceeds as follows. In Section 2 we outline the GEL problem in some detail. We then outline the arguments under which the pseudo-true value exists in Section 3. In Section 4 we develop the asymptotic theory. Simulation results are in Section 5, and then we finally conclude.

2 Generalized Empirical Likelihood Problem

EL, as popularized in Owen (1988), Qin and Lawless (1994), and Owen (2001), uses a nonparametric empirical analogue of the classic maximum likelihood framework to estimate a parameter θ in moment models. In the framework discussed below, EL penalises adjusting the weight placed on each observation relative to the empirical distribution (weight of $1/n$, n is sample size) using log. In Smith (1997) and later in Newey and Smith (2004), alternatives to log are discussed. Here we will consider a subset of the frequently used Cressie-Read family of divergences that includes EL as a special case.

Briefly, given a sample of observations $X_1, \ldots, X_n$ on a sample space $\mathcal{X}$, the goal is to find weights $\pi_1, \ldots, \pi_n \in \mathbb{R}^n$ and estimator that solve the following problem:

$$\min_{\theta,\pi} \frac{1}{n} \sum_{i=1}^{n} \varphi\left(\frac{\pi_i}{1/n}\right) \tag{1}$$

$$s.t. \quad \sum_{i=1}^{n} \pi_i = 1 \tag{2}$$

$$\frac{1}{n} \sum_{i=1}^{n} \pi_i g(X_i, \theta) = 0. \tag{3}$$

where $g : \mathbb{X} \times \Theta \to \mathbb{R}^q$ are the moment conditions representing the econometric model, and φ is of the form

$$\varphi(x) = \frac{x^{\gamma+1} - 1 - (\gamma+1)(x-1)}{\gamma(\gamma+1)}, \quad \gamma < 0. \tag{4}$$

EL corresponds to the case where $\gamma \to -1$. Other common values are Hellinger Distance ($\gamma = -1/2$) and Pearson χ^2 ($\gamma = -2$). We focus on this case because the Cressie-Read family is the most commonly used family for GEL estimation and in Schennach (2007) it was shown that for these choices of φ, for distributions with unbounded support, under misspecification there is no pseudo-true parameter which can be estimated at the $\sqrt{n}$-rate.

We will refer to the problem in (1) as the GEL Empirical Primal Problem (EPP). For fixed θ, we will let $\hat{\Pi}(\theta) = \{\pi : \sum_{i=1}^n \pi_i = 1, \frac{1}{n}\sum_{i=1}^n \pi g(X_i, \theta) = 0\}$ be the feasible set of weights. We will refer sometimes to the value of the problem for fixed θ, which is defined as $V_{EPP}(\theta) = \min_{\pi \in \hat{\Pi}(\theta)} \frac{1}{n}\sum_{i=1}^n \varphi(\pi_i n)$. The population analogue of the EPP is what we will refer to as the Primal Problem (PP):

$$\inf_{\theta, f} \int_{\mathcal{X}} \varphi(f(x))P(dx) \tag{5}$$

$$s.t. \int_{\mathcal{X}} f(x)P(dx) = 1 \tag{6}$$

$$\int_{\mathcal{X}} g(x,\theta)f(x)P(dx) = 0. \tag{7}$$

At the population level, we optimise over densities f with respect to P. The value of this problem in (5) is $V_{PP}(\theta)$. Let $\Pi(\theta)$ be defined in a similar fashion to $\hat{\Pi}(\theta)$:

$$\Pi(\theta) := \left\{ f : \int_{\mathcal{X}} f(x)P(dx) = 1, \quad \int_{\mathcal{X}} g(x,\theta)f(x)P(dx) = 0 \right\} \tag{8}$$

The set $\Pi(\theta)$ is clearly a convex set.

There are two particularly important features of φ of the type (4) when $\gamma < 0$: the behaviour of φ' near 0, and behaviour of $\varphi(x)/x$ as $x \to \infty$. First, note that $\varphi'(0) = -\infty$. This implies that for this range of γ, P must also be absolutely continuous with respect to $f_\theta dP$. That is, if f_θ is optimal, it must be the case that if $\int_A f_\theta dP = 0$, then $P(A) = 0$. For the EPP, this implies that all weights will be strictly positive.

Now, consider that $\lim_{u\to\infty} \varphi(u)/u < \infty$. This implies that we do not penalise much for distributions which place high probability on sets where P assigns zero probability. This means we must look at solutions in this case where there is mutual absolute continuity. The property $\lim_{u\to\infty} \varphi(u)/u < \infty$ is a lack of coercivity of the objective, and will imply that we will need to consider extended solutions in general.

Solving the EPP in equations (1)-(3) is typically done using duality. Recall that the convex conjugate f^* of a function $f : X \to \mathbb{R}$ is defined as

$$f^*(y) := \sup_{x \in X}\{\langle x, y\rangle - f(x)\} \tag{9}$$

where clearly we must allow that f^* can take on the value $+\infty$. The domain of f^*, $\operatorname{dom} f^*$, is the set of y such that $f^*(y) < \infty$. The importance of the conjugate is in duality theory. For the EPP, the empirical dual problem (EDP) is then

$$\min_{\theta} \max_{\eta \in \mathbb{R}, \lambda \in \mathbb{R}^q} \left\{\eta - \frac{1}{n}\sum_{i=1}^{n} \varphi^*\left(\eta + \lambda' g(X_i, \theta)\right)\right\} \tag{10}$$

subject to the condition that $\eta + \lambda' g(X_i, \theta) \in \operatorname{dom} \varphi^*$ for all i. We will refer to this set as $\hat{\Lambda}(\theta) = \{\eta, \lambda : \eta + \lambda' g(X_i, \theta) \in \operatorname{dom} \varphi^*\}$. This constraint is linear in the control variables η and λ, and therefore the dual forms a convex programme for fixed θ. The dual problem EDP can be related to the primal problem EPP for each θ. We will similarly define the value of the dual problem as $V_{EDP}(\theta) = \max_{(\eta,\lambda) \in \hat{\Lambda}(\theta)}\{\eta - \frac{1}{n}\sum_{i=1}^{n} \varphi^*(\eta + \lambda' g(X_i, \theta))\}$. Under certain conditions on φ, a strong duality result will give us that

$$V_{EDP}(\theta) = V_{EPP}(\theta) \tag{11}$$

This has historically been an attractive result from a practical perspective. If this holds, then the estimator of θ is

$$\hat{\theta} := \min_{\theta} V_{EDP}(\theta). \tag{12}$$

We will assume that $\Theta \subset \mathbb{R}^p$. Thus, this is an optimization problem in $p+q+1$ variables, whereas the primal problem EPP is an optimization problem in $n+p$ variables. In fact, in many places the GEL estimator has been defined in this way, skipping the primal problem altogether.

Similarly to the primal problem, we can also define a population dual problem,

$$\inf_{\theta} \sup_{\eta,\lambda} \left\{\eta - \int \varphi^*(\eta + \lambda' g(x, \theta)) P(dx)\right\}, \tag{13}$$

which we will refer to as DP. The relationship between PP and DP has been extensively discussed in Rockafellar (1968), Rockafellar (1971), and Borwein and Lewis (1991), to name a few. When achieved, we will refer to the value of the problem as $V_{DP}(\theta) := \sup_{(\eta,\lambda)\in\Lambda(\theta)} \eta - \int_{\mathcal{X}} \varphi^*(\eta + \lambda' g(x, \theta)) P(dx)$.

We mentioned above that the inner sup in (13) has a side condition. For the class of φ^* we consider, they are all of the form

$$\varphi^*(v) = \frac{(1+\gamma v)^{\frac{\gamma+1}{\gamma}} - 1}{\gamma+1}. \tag{14}$$

The relevant range is for $v < -1/\gamma$. Thus, it is helpful to note that we can precisely define $\Lambda(\theta)$, the population counterpart of $\hat{\Lambda}(\theta)$ in these cases: $\{\eta, \lambda : \eta + \lambda' g(X_i, \theta) < -1/\gamma \,\text{w.p.}\, 1\}$. For compact $\mathcal{X}$ and bounded g this places constraints on the Lagrange multipliers. If the Lagrange multipliers end up on the boundary, then we may not have primal attainment, and thus we may have a gap $V_{PP}(\theta) - V_{DP}(\theta) > 0$.

3 Existence of Pseudo-true values

In the previous section, the GEL problem was reviewed. In this section we review some results from the convex analysis and information theory literatures, and give sufficient conditions for the existence of the pseudo-true value θ_0 that has a natural projection interpretation.

We will begin stating our key assumptions. The first assumptions are on the moment conditions $g(\cdot, \theta)$ and the sample space $\mathcal{X}$.

Assumption 3.1. *The sample space $\mathcal{X}$ and parameter space Θ are compact, $g(X_i, \theta)$ is twice-continuously differentiable in θ on the interior of Θ, and $\sup_{\theta \in \Theta} \sup_{x \in \mathcal{X}} \|g(x, \theta)\| < \infty$. Furthermore, for all θ, $0 \in g(\mathcal{X}, \theta)$ and $\inf_\theta \lambda_{min}(\mathbb{E}\, g(X, \theta) g(X, \theta)') > 0$.*

The compactness of $\mathcal{X}$ plus boundedness of $g(x, \theta)$ ensure that for each θ, $V_{PP}(\theta)$ is finite. The smoothness assumption greatly simplifies the exposition of the main results while still highlighting the important divergences from the existing theory. The boundedness assumption is also important for solution existence and uniqueness. The assumption that $g(x, \theta)$ is uniformly bounded for each θ implies that the set $\Pi(\theta)$ is also closed. This assumption was investigated thoroughly in Schennach (2007), where it was shown that for EL, if the moment conditions are unbounded then under misspecification, the EL-estimator cannot be $\sqrt{n}$-consistent for any parameter value in the parameter space. We have therefore ruled out the issues with misspecification discussed in that paper. The last assumption ensures that for any θ, there is a feasible solution to PP.

The interesting feature here is that in general the population dual problem DP is still finite-dimensional, where the population primal problem PP is infinite dimensional. There are some key differences, however, which lead to the statistical challenges that are at the core of this paper. The main difference is that while the dual of EDP is in turn EPP, the dual of DP is in general not PP. While it may be true that the biconjugate of φ is again φ (i.e. $(\varphi^*)^* = \varphi$) even in that case it is not generally true that the dual of DP is again PP. Consider the integral functional in the primal problem as a functional $I_{\theta,\varphi} : L^1(P) \to (-\infty, \infty]$, where we use the standard notation that $L^1(P)$ is the set of P-integrable functions and the inclusion of $+\infty$ in interval notation implies that $I_{\theta,\varphi}$ is extended-real valued. The dual-space of $L^1(P)$ is $L^\infty(P)$, the space of almost-everywhere-P bounded functions, and therefore the dual objective function is another integral functional, this time $I_{\theta,\varphi^*} : L^\infty(P) \to [-\infty, \infty)$, where the element in $L^\infty(P)$ is indexed by η and λ. Now, when finding the dual of I_{θ,φ^*}, the following issue must be accounted for: the dual space of $L^\infty(P)$ is not $L^1(P)$. Therefore, the dual of DP augments the original problem with a component in the space of signed measures. The extended primal problem (PP*) will be denoted

$$\inf_{\theta,f,\nu} \int_{\mathcal{X}} \varphi(f(x))P(dx) + q\nu^+(\mathcal{X}) - p\nu^-(\mathcal{X}) \tag{15}$$

$$s.t. \int_{\mathcal{X}} f(x)P(dx) + \nu(\mathcal{X}) = 1 \tag{16}$$

$$\int_{\mathcal{X}} g(x,\theta)f(x)P(dx) + \int_{\mathcal{X}} g(x,\theta)\nu(dx) = 0 \tag{17}$$

where $q = \lim_{x\to\infty} \varphi(x)/x$, $p = \lim_{x\to-\infty} \varphi(x)/x$ and ν is singular with respect to P, i.e. $P \perp \nu$. See Borwein and Lewis (1993) for a full discussion. Note that for the choices of φ we consider, $p = -\infty$ in our setting and therefore in any solution $\nu^-(\mathcal{X}) = 0$. In what follows we will therefore ignore the decomposition of ν into positive and negative parts and retain only ν. We will return to this point below briefly when discussing estimation. As before, we denote the achieved infimum over the constraint set in (15), for each θ, by $V_{PP^*}(\theta)$.

It turns out that this second problem PP* is not merely a theoretical construction: solutions to PP* can be interpreted as projections. For fixed θ, if we define a measure $\mu_\theta \in \Pi(\theta)$ by $\mu_\theta(A) = \int_A f_\theta dP + \nu_\theta(A)$, where f_θ and ν_θ are optimal for θ, then μ_θ defines the closest distribution to P under which the moment conditions are mean-zero, where 'closest' is determined by φ. Let

$$\Pi^*(\theta) = \left\{ \mu : \mu(\mathcal{X}) = 1, \quad \int_{\mathcal{X}} g(x,\theta)\mu(dx) = 0 \right\}. \tag{18}$$

Then, the solution to PP* can be viewed as the φ-projection of P onto $\Pi^*(\theta)$. In general, f_θ can be interpreted as the generalized projection of P onto the set $\Pi(\theta)$: any sequence of functions f_n such that $\int_{\mathcal{X}} \varphi(f_n)dP \to \int_{\mathcal{X}} \varphi(f_\theta)dP + q\nu_\theta(\mathcal{X})$ must converge to f_θ. The 'generalized' part comes from the fact that there is no guarantee that

$f_\theta \in \Pi(\theta)$ whenever $\nu_\theta(\mathcal{X}) > 0$. For more discussion of φ-projections and generalized projections, see Csiszár (1995).

Now, we proceed to our first result. The parameter θ has been present in our discussion up until now, however we have largely focused on summarizing the setup for fixed θ where the relevant results already exist in the convex analysis literature. Our first result gives conditions under which a pair (θ, μ) exist that solve PP* and when that solution can be found from DP.

Proposition 1. *Assume that there exists a $\theta \in \Theta$ such that $\mathbb{E}\, g(X_i, \theta)$ is in the convex hull of the range of $g(\cdot, \theta)$ on $\mathcal{X}$. Under assumptions 3.1 with φ^* of the form (4), the set*

$$\arg\min_\theta V_{PP^*}(\theta) \tag{19}$$

is nonempty. Furthermore, $V_{DP}(\theta) = V_{PP^*}(\theta)$, and thus it also holds that

$$\arg\min_\theta V_{PP^*}(\theta) = \arg\max_\theta V_{DP}(\theta) \tag{20}$$

The above theorem states that a θ which minimizes (PP*) exists, and it can be found by solving the dual problem DP. For fixed θ, this is simply an application of Theorems 3.4 and 4.1 in Borwein and Lewis (1993); see also Csiszár (1995) and Teboulle and Vajda (1993). Compactness of $\mathcal{X}$ and boundedness of the maps $g(\cdot, \theta)$ imply that the necessary dual constraint qualification holds, while the convex hull assumption implies that there is at least one θ such that $V_{PP^*}(\theta)$ exists. With the assumption on strict convexity of φ, the absolutely-continuous part f_θ of any solution will also be unique.

Without stronger assumptions, however, we cannot say that the same holds true of PP. This generally will occur for a given θ when the solution $(\eta_\theta, \lambda_\theta)$ for the dual problem occurs on the boundary of $\Lambda(\theta)$. In these cases, the solution to the dual problem might no longer be feasible in PP. It is also common to 'concentrate out' the parameter η in the dual problem and instead consider solutions to

$$\inf_\lambda \int_{\mathcal{X}} \varphi^*(1 + \lambda' g(x, \theta)) P(dx) \tag{21}$$

and then take the solution in the primal problem to be

$$\tilde{f}_\theta(x) = \frac{\varphi^{*\prime}(1 + \lambda' g(x, \theta))}{\int_{\mathcal{X}} \varphi^{*\prime}(1 + \lambda' g(x, \theta)) P(dx)} \tag{22}$$

This is the solution when the solution to the dual problem

$$f_\theta(x) = \varphi^{*\prime}(\eta + \lambda' g(x, \theta)) \tag{23}$$

is feasible for PP. In general, when f_θ is not feasible for PP, $\tilde{f}_\theta \neq f_\theta$, and θ resulting from finding the minimum of (21) over θ cannot be interpreted as a parameter defined by a projection in the same sense that a minimizing θ of (19) can be. Furthermore, in cases where the implied probabilities $\hat{\pi}$ from the solution of EPP are of interest, the resulting normalised probabilities will not satisfy the moment conditions. This makes this renormalisation unsuitable for reweighting a sample to match another population, as in Hellerstein and Imbens (1999). They also will no longer be useful for obtaining higher-order refinements of the bootstrap, as in Brown and Newey (2002).

The new part of the above result is the dependence on θ which requires a couple of comments. The proof relies on an implicit function theorem for the solution mapping in the dual problem, $(\eta, \lambda)(\theta) : \Theta \to \mathbb{R}^{q+1}$. Convexity of φ^* is used to show that this mapping must be at least semidifferentiable in θ, and therefore continuous in θ.

Our result contrasts and complements the existence results in Schennach and Wahlstrom (2024). The focus in that paper is existence under bounded moment conditions of the pseudo-true parameter in the dual problem. We instead focus on existence of the pseudo-true parameter in the primal problem, defined as a generalised projection. Rather than focusing on whether there exists any pseudo-true value, we focus our attention on the question of existence of a particular pseudo-true value.

4 Asymptotic Theory

The discussion in the previous section is relevant in econometric practice when $P \notin \Pi^*(\theta)$ for all θ. Under correct specification, the unique minimizer is in fact the distribution P. Thus, we concern ourselves with the case where $\mathbb{E}\, g(X_i, \theta) \neq 0$ for all $\theta \in \Theta$.

A natural starting point is to find $\hat{\theta}$ by minimizing $V_{EDP}(\theta)$, that is

$$\hat{\theta} = \arg\min_{\theta} \max_{(\eta,\lambda)\in\hat{\Lambda}(\theta)} \left\{ \eta - \frac{1}{n} \sum_{i=1}^{n} \varphi^*(\eta + \lambda' g(X_i, \theta)) \right\}. \tag{24}$$

We will now assume that there is a unique pseudo-true value:

Assumption 4.1. *The data $X_1, \ldots, X_n$ are i.i.d. There is a unique $\theta_0 \in \Theta$ with $\theta_0 = \arg\min_\theta V_{PP*}(\theta)$.*

We make the uniqueness assumption primarily because questions of partial-identification in these projection settings, while interesting, are outside the scope of this paper. The i.i.d. assumption can be replaced by a variety of assumptions which will guarantee the law of large numbers and central limit theorem below. With a target parameter in hand, we now turn to consistency of the estimator and characterizing its limit distribution.

4.1 Consistency

We first show that $\hat{\theta} \xrightarrow{P} \theta_0$. The primary wrinkle here is that $\hat{\theta}$ involves an inner estimation of $(\eta, \lambda)(\theta)$: we can write our estimator as

$$\hat{\theta} = \arg\min_{\theta} \left\{ \hat{\eta}(\theta) - \frac{1}{n} \sum_{i=1}^{n} \varphi^*(\hat{\eta}(\theta) + \hat{\lambda}(\theta) g(X_i, \theta)) \right\}, \tag{25}$$

$$(\hat{\eta}, \hat{\lambda})(\theta) := \arg\max_{\eta, \lambda} \left\{ \eta - \frac{1}{n} \sum_{i=1}^{n} \varphi^*(\eta + \lambda' g(X_i, \theta)) \right\} \tag{26}$$

The infeasible estimator we would like to implement does not involve the empirical estimates $(\hat{\eta}, \hat{\lambda})(\theta)$; rather, we would like to use

$$\hat{\theta} = \arg\min_{\theta} \left\{ \eta(\theta) - \frac{1}{n} \sum_{i=1}^{n} \varphi^*(\eta(\theta) + \lambda(\theta) g(X_i, \theta)) \right\}, \tag{27}$$

$$(\eta, \lambda)(\theta) := \arg\max_{\eta, \lambda} \left\{ \eta - \int_{\mathcal{X}} \varphi^*(\eta + \lambda' g(x, \theta)) P(dx) \right\} \tag{28}$$

Attaining consistency is straightforward.

Theorem 1. *For φ in the CR family, under assumptions 3.1 and 4.1,*

$$\hat{\theta} \xrightarrow{P} \theta_0 \tag{29}$$

4.2 Asymptotic Distribution

For the asymptotic distribution theory, two considerations must be made. In the CR family, we must allow for corner solutions for (η, λ) when $\gamma < 0$.

Theorem 2. *Let $X_1, \ldots, X_n$ be an i.i.d. sample from P. For φ in the CR family with* *parameter $\gamma < 0$, and under assumptions 3.1 and 4.1, we have that*

$$\sqrt{n}(\hat{\theta} - \theta_0) \Rightarrow \arg\min_s \left\{ -\partial(\eta, \lambda)(\theta_0)(s)' Z_{\theta_0} - \frac{1}{2} \partial(\eta, \lambda)(\theta_0)(s)' \Omega(\theta_0) \partial(\eta, \lambda)(\theta_0)(s) \right\} \tag{30}$$

where

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \begin{pmatrix} \varphi^{*\prime}(\eta(\theta_0) + \lambda(\theta_0)' g(X_i, \theta_0)) - 1 \\ \varphi^{*\prime}(\eta(\theta_0) + \lambda(\theta_0)' g(X_i, \theta_0)) g(X_i, \theta_0) \end{pmatrix} \Rightarrow Z_{\theta_0} \tag{31}$$

$$\Omega(\theta_0) = \mathbb{E}\, g(X_i, \theta_0) g(X_i, \theta_0)'. \tag{32}$$

The proof leverages the convexity of φ^* and $\varphi^{*\prime}$ along with the uniform convergence established in Theorem 1, but otherwise is a straightforward application of Theorem 6 in Andrews (1999).

The boundary dependence goes through $\partial(\eta, \lambda)$. In cases where changing θ does not change η or λ because one or both are on the boundary, then we have superconsistency of $\hat{\theta}$ and $\sqrt{n}(\hat{\theta} - \theta_0) \to 0$ in probability.

We do not address inference in this paper. The limit distribution could be computed via Monte Carlo as in Dufour (2006). We do not recommend the bootstrap here, however the subsampling method of Politis et al. (2001) could be used to obtain asymptotically valid confidence intervals, which would have uniform validity regardless of whether the limit distribution was Gaussian or not.

5 Simulations

We motivate our simulations with the setup in Schennach (2007), modified to fit our bounded $g(X_i, \theta)$ setting and matchup with an example in Borwein and Lewis (1993) where we have analytic solutions for EL, even in the extended problem. Our moment conditions are

$$g(X_i, \theta) = \begin{pmatrix} X_i - \theta \\ \sqrt{X_i} - \alpha \end{pmatrix} \tag{33}$$

where α is a fixed constant. We will consider sample sizes of 500 and 5000. For α we will consider 0.10, 0.25, 0.50, 0.66, and 0.9. The true distribution of the data will be $X_1, \ldots, X_n$ i.i.d. Uniform on the unit interval $[0, 1]$. Under this data generating process, $\alpha = 0.66$ is the only correctly-specified model. We will compare the extended GEL procedure recommended here to the continuously updating (CU) estimator and the traditional EL, Hellinger Distance (HD), Exponential Tilting (ET), and chi-square

(CS) estimators. In all cases we use 5000 simulations.

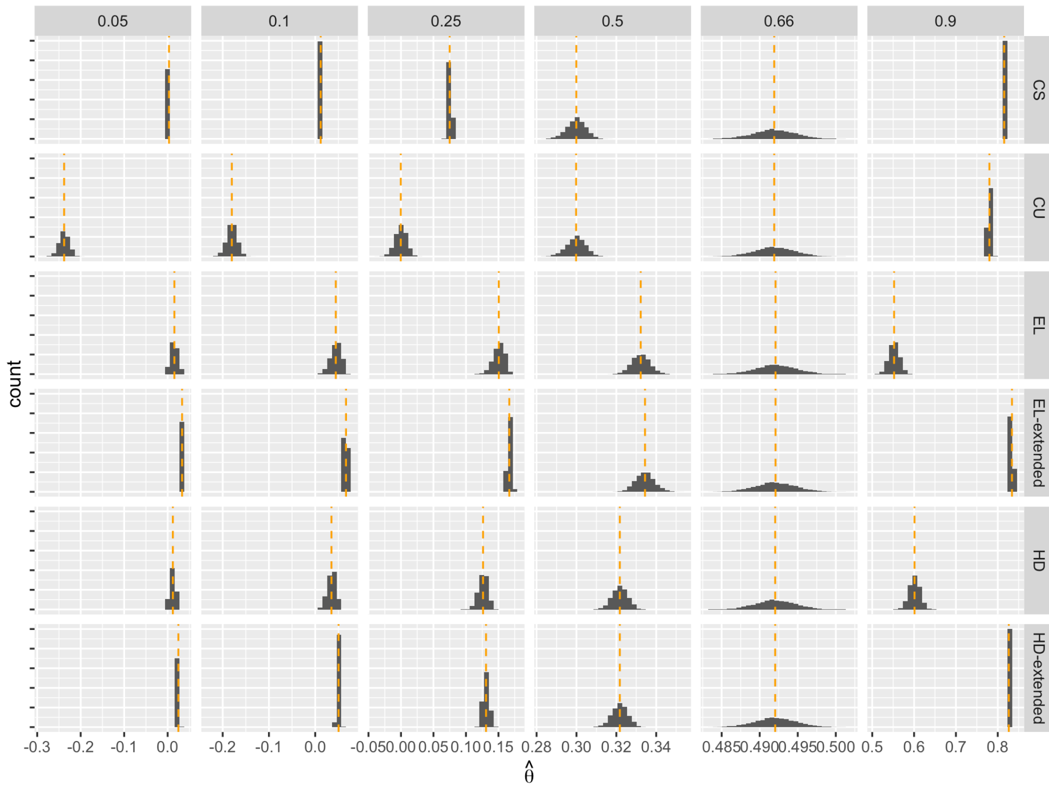

Figure 1: Sample size of 500. α changes across columns, the estimator used across rows. The pseudo-true value is the dashed orange line.

We first look at histograms of the parameter estimates, $\hat{\theta}$, in Figures 1 and 2. Notice that the x-axis is different for each α. As α increases, the pseudo-true values increase as well. Across estimators there is much variation. Notice that for $\alpha = 0.66$, correct specification, all the procedures we consider lead to the same pseudo-true value and the same distribution, as is expected. Comparing the CU and CS estimators, under mild misspecification they essentially agree; few observations are assigned negative weights under CU or zero weight under CS. Under larger degrees of misspecification, notice that the pseudo-true value for CU exists the support of the data, while the CS estimator becomes increasingly concentrated, as much of the support is given zero weight to match the moment conditions. EL and our version, EL-extended, are quite similar for low α, but interestingly the pseudo-true values are quite different for large alpha. Similar findings hold for HD.

Next, we look at the density plots of the rescaled estimates, $\sqrt{n}V_{MC}^{-1/2}(\hat{\theta}-\theta)$, where V_{MC} is a Monte-Carlo estimate of the variance of $\hat{\theta}$. Under correct specification the sampling distribution of all of the estimators look to be well approximated by the standard normal. In general, it is only for small values of α where the deviations from

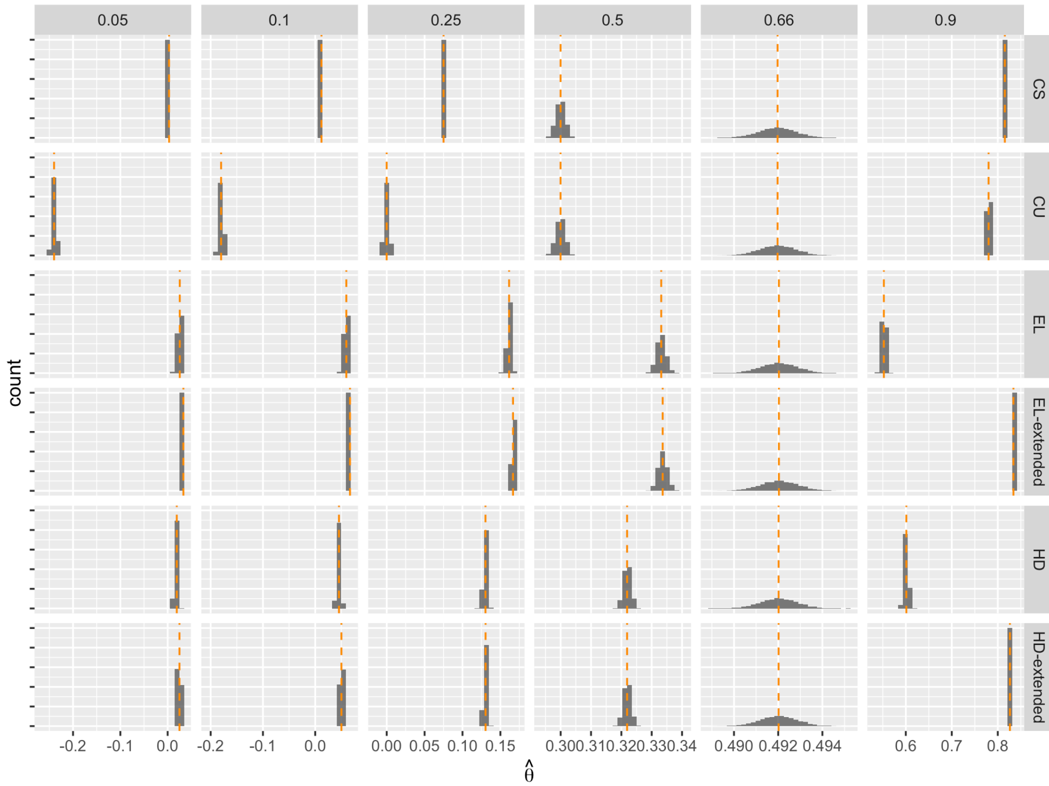

Figure 2: Sample size of 5000. α changes across columns, the estimator used across rows. The pseudo-true value is the dashed orange line.

normality begin. Notice that the CU estimator is always asymptotically normal; the extrapolation and smoothness of the criterion means that under misspecification the estimator is still asymptotically normal. The CS estimator is poorly approximated by its limit when n is small, however note that this is a small-sample issue. For a sample size of 5000 the normal approximation seems sufficient. EL, EL-extended, and the HD counterparts all exhibit departures from normality for small α. $(\eta, \lambda)(\theta)$ are on the boundary of the domain of φ^* in these cases, leading to some irregularity in the distribution of $\hat{\theta}$.

These findings are supported by the quantile-quantile plots in Figures 5 and 6, where the standard normal quantiles are plotted against the quantiles of the standardized estimators. As previously discussed, CU is in generally quite well approximated by a normal distribution. It is worth noting that here it becomes clear that the traditional EL and HD estimators exhibit much starker departures from normality than the extensions discussed in this paper. Indeed, the sampling distribution of the version of EL we develop here is well approximated by a normal distribution for all but the cases with the most severe misspecification. The same cannot be said for the more traditional EL, which exhibits departures from normality beginning when $\alpha = 0.25$.

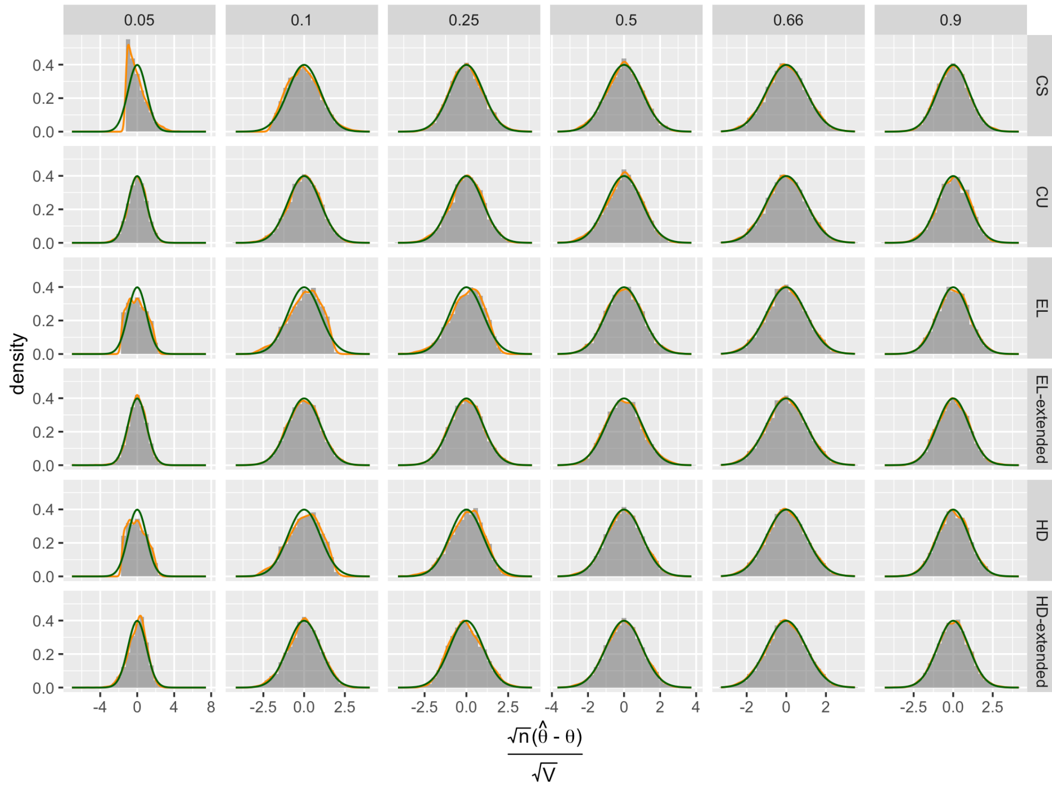

Figure 3: Sample size of 500. α changes across columns, the estimator used across rows. The orange density is fitted from the simulations, the green density is the standard normal.

6 Conclusion

In this paper we clarify how we should interpret the pseudo-true values in GEL estimation problems. The pseudo-true values can be considered generalized projections of the data generating process onto the set of probability distributions which satisfy the moment conditions. Since standard estimation approaches will potentially fail to have this interpretation, we developed modified procedures which are consistent for the projection pseudo-true values. The estimators will not necessarily be asymptotically normal, however in simulations it appears the distortions are not necessarily severe unless the degree of misspecification is large.

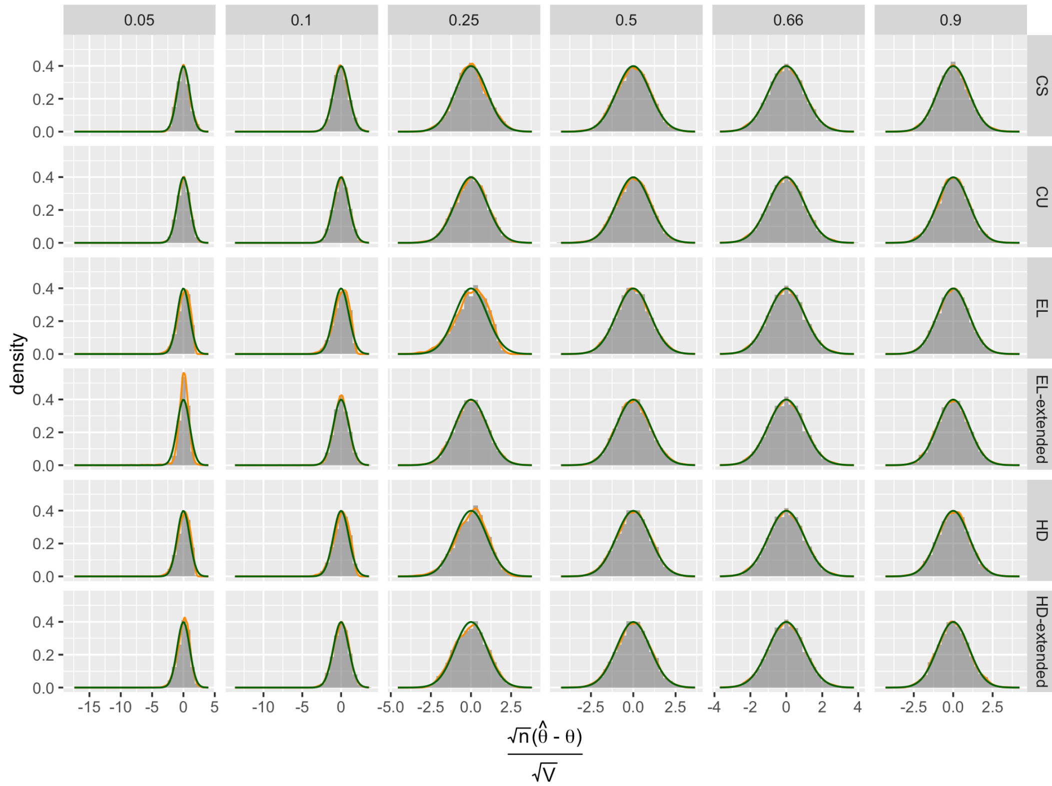

Figure 4: Sample size of 5000. α changes across columns, the estimator used across rows. The orange density is fitted from the simulations, the green density is the standard normal.

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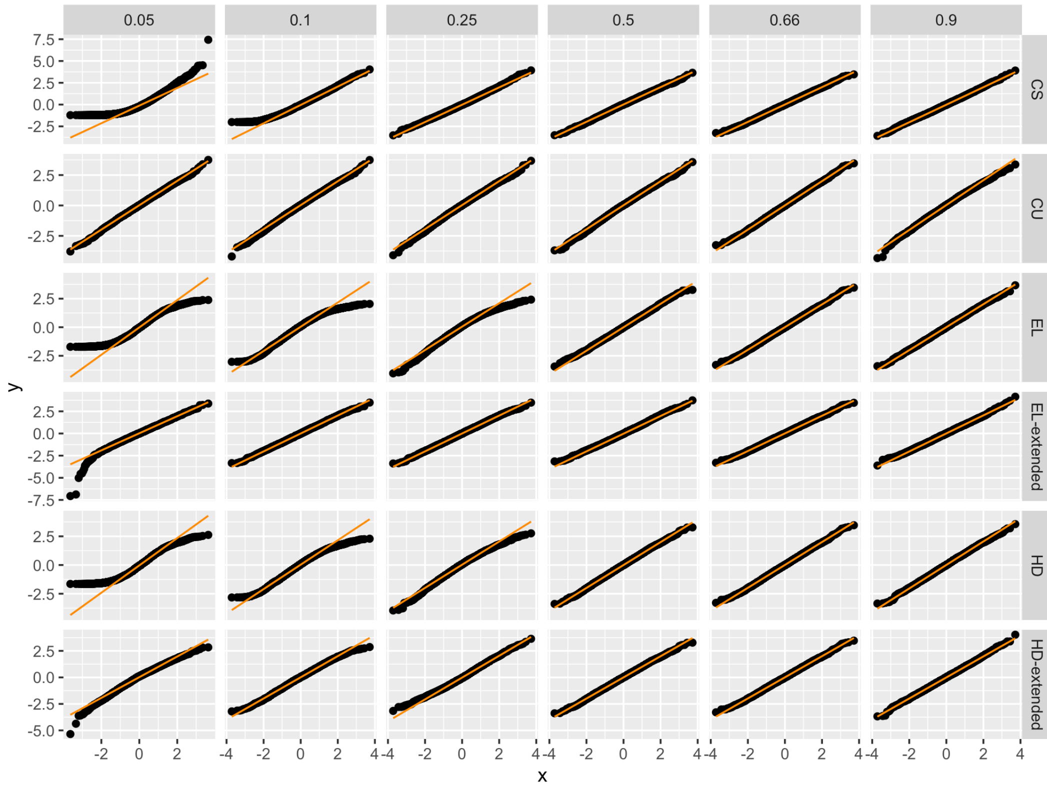

Figure 5: Sample size of 500. α changes across columns, the estimator used across rows. The orange line is the 45-degree line, normal quantiles are plotted against quantiles of the standardized estimator.

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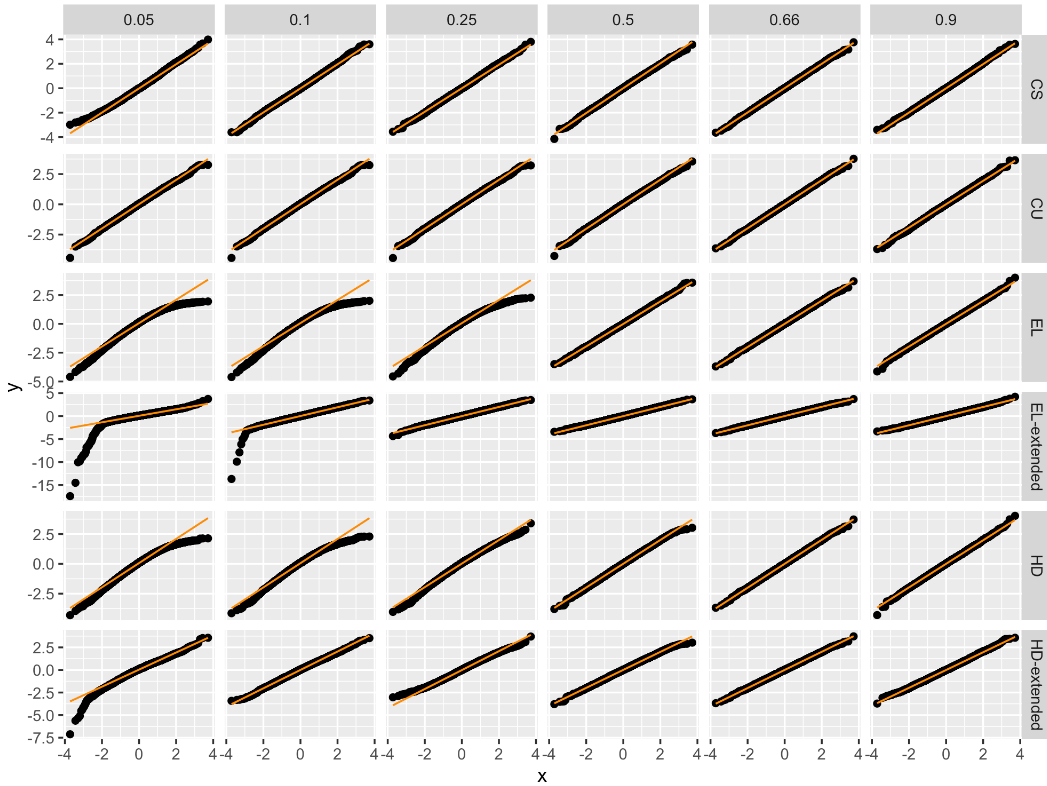

Figure 6: Sample size of 5000. α changes across columns, the estimator used across rows. The orange line is the 45-degree line, normal quantiles are plotted against quantiles of the standardized estimator.

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A Proofs

A.1 Proof of Theorem 1

Proof. Since Θ is compact, the sample space is compact, and $g(x, \theta)$ is bounded and continuous, the induced parameter space for η, λ is also compact by the implicit function theorem. This comes from the identification condition that $\inf_\theta \lambda_{min}(\mathbb{E}\, g(X,\theta)g(X,\theta)') > 0$, as well as the fact that for CR divergences with $\gamma > 0$, $\varphi^{*\prime\prime} > 0$, and thus as the Jacobian for the implicit function theorem is

$$\begin{pmatrix} \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) & \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta)' \\ \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta) & \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta) g(X,\theta)' \end{pmatrix} \tag{34}$$

we have that $\eta(\theta)$ and $\lambda(\theta)$ are continuous. Thus, the set of potential η, λ is compact. This implies that the sequence of estimators $\hat{\eta}, \hat{\lambda}$ is uniformly tight, as the Jacobian for the empirical problem will be invertible with probability approaching one, and thus $\hat{\eta}, \hat{\lambda}$ will be in the same compact set with high probability. We can then apply Corollary 3.2.3 from Vaart and Wellner (2023): the set of functions $\varphi^*(\eta(\theta) + \lambda(\theta)' g(X,\theta))$ is Lipschitz in θ, and therefore the necessary uniform convergence on compact subsets $K \subset \Theta$ holds. $\square$

A.2 Proof of Theorem 2

Proof. Using the implicit function theorem, we build a quadratic approximation: the main difference is that the derivatives of η and λ are not guaranteed to be full-rank, therefore we cannot assume we can solve for the minimiser of the approximating expansion explicitly. The implicit function theorem gives us derivatives

$$-\begin{pmatrix} \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) & \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta)' \\ \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta) & \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta) g(X,\theta)' \end{pmatrix}^{-1} \tag{35}$$

$$\begin{pmatrix} \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) \lambda' \partial_\theta g(X,\theta) \\ \mathbb{E}\, \varphi^{*\prime\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) g(X,\theta) \lambda(\theta)' \partial_\theta g(X,\theta) + \varphi^{*\prime}(\eta(\theta) + \lambda(\theta)' g(X,\theta)) \partial_\theta g(X,\theta) \end{pmatrix}. \tag{36}$$

We then have the quadratic expansion to apply Theorem 3 in Andrews (1999) and the result follows. $\square$